



Introduction to the Program Design and Programming

ICT 1305

CHAPTER 04

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What is a Flowchart

- A flowchart is a visual representation of the sequence of steps and decisions needed to perform a process.
- Each step in the sequence is noted within a diagram shape.
- Steps are linked by connecting lines and directional arrows.
- This allows anyone to view the flowchart and logically follow the process from beginning to end.
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What is a Flowchart

flowcharts can:

- Demonstrate the way code is organized.
- Visualize the execution of code within a program.
- Show the structure of a website or application.
- Understand how users navigate a website or program.

Basic Flowchart Symbols

Symbol	Name	Function
	Start/end	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectangle represents a process
	Decision	A diamond indicates a decision

Flowchart Symbols

Start/End Symbol

The terminator symbol marks the starting or ending point of the system. It usually contains the word "Start" or "End."



Flowchart Symbols

Action or Process Symbol

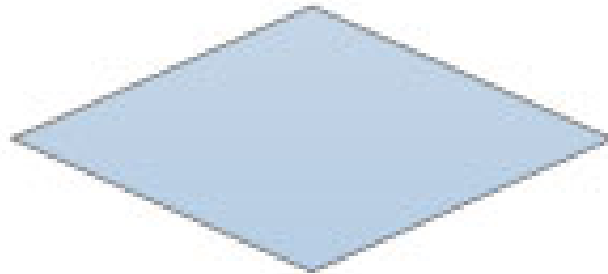
A box can represent a single step ("add two cups of flour"), or an entire sub-process ("make bread") within a larger process.



Flowchart Symbols

Decision Symbol

A decision or branching point. Lines representing different decisions emerge from different points of the diamond.



Flowchart Symbols

Input/Output Symbol

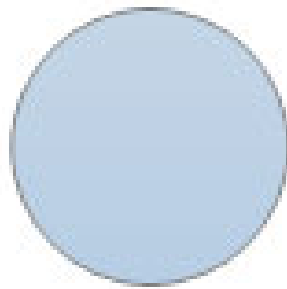
Represents material or information entering or leaving the system, such as customer order (input) or a product (output).



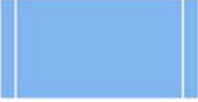
Flowchart Symbols

Connector Symbol

Indicates that the flow continues where a matching symbol (containing the same letter) has been placed.

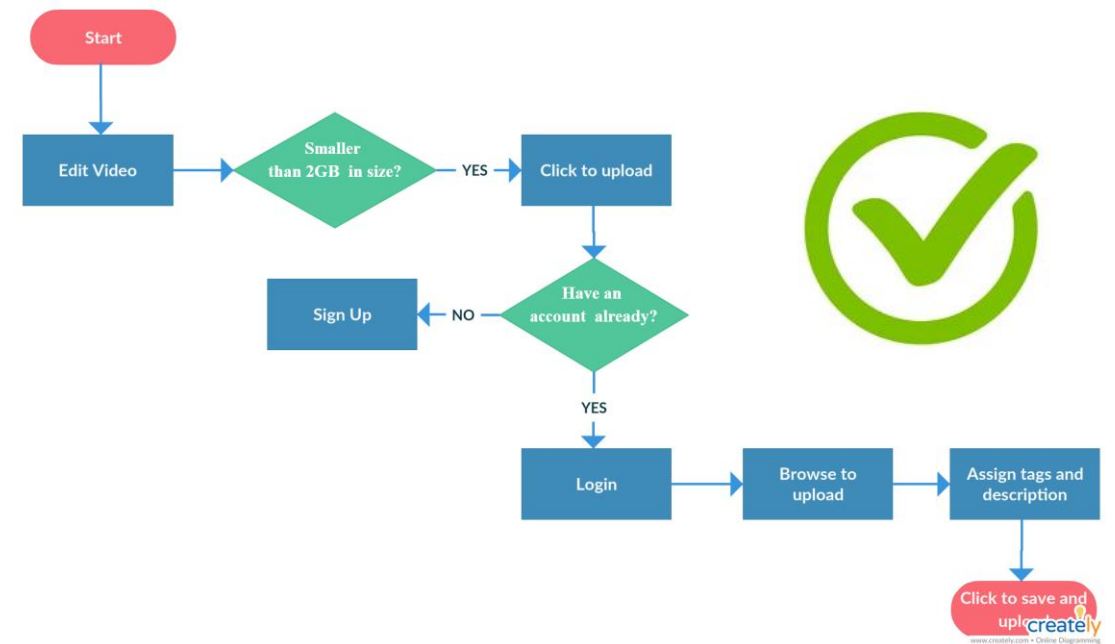
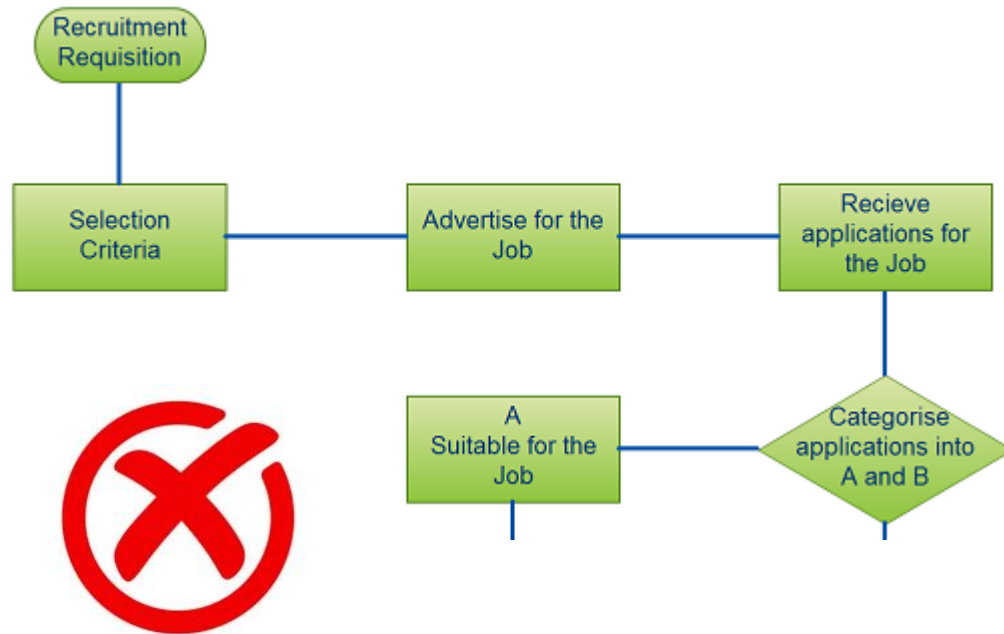


Other Flowchart Symbols

	Terminator Indicates the beginning or end of a program flow in your diagram.		Subroutine Indicates a predefined (named) process, such as a subroutine or a module.		Connector Indicates an inspection point.		Collate Indicates a step that organizes data into a standard format.
	Process Indicates any processing function.		Preparation Indicates a modification to a process, such as setting a switch or initializing a routine.		Off-page connector Use this shape to create a cross-reference and hyperlink from a process on one page to a process on another page.		Sort Indicates a step that organizes items list sequentially.
	Decision Indicates a decision point between two or more paths in a flowchart.		Display Indicates data that is displayed for people to read, such as data on a monitor or projector screen.		Off-page connector		Merge Indicates a step that combines multiple sets into one.
	Delay Indicates a delay in the process.		Manual input Indicates any operation that is performed manually (by a person).		Off-page connector		Database Indicates a list of information with a standard structure that allows for searching and sorting.
	Data Can represents any type of data in a flowchart.		Manual loop Indicates a sequence of commands that will continue to repeat until stopped manually.		Off-page connector		Internal storage Indicates an internal storage device.
	Document Indicates data that can be read by people, such as printed output.		Loop limit Indicates the start of a loop. Flip the shape vertically to indicate the end of a loop.		Or Logical OR		
	Multiple documents Indicates multiple documents.		Stored data Indicates any type of stored data.		Summing junction Logical AND		

General Rules in Flowcharting

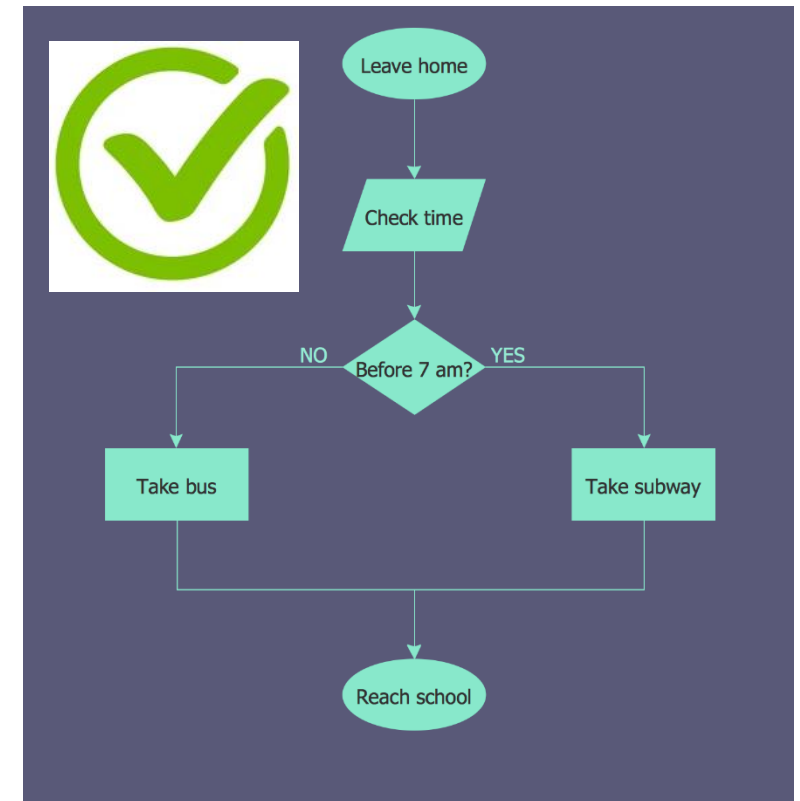
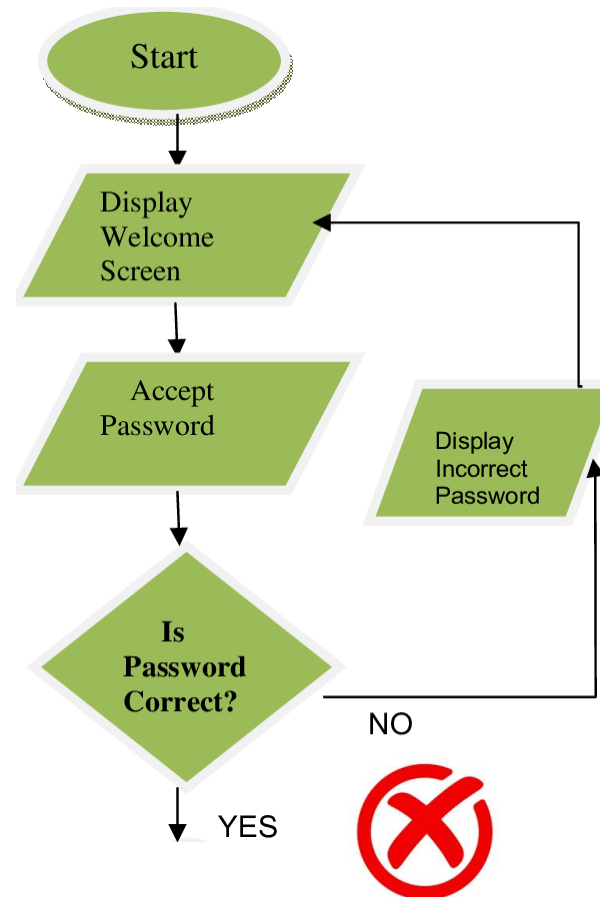
1. All boxes of the flowchart are connected with Arrows. (Not lines)



General Rules in Flowcharting

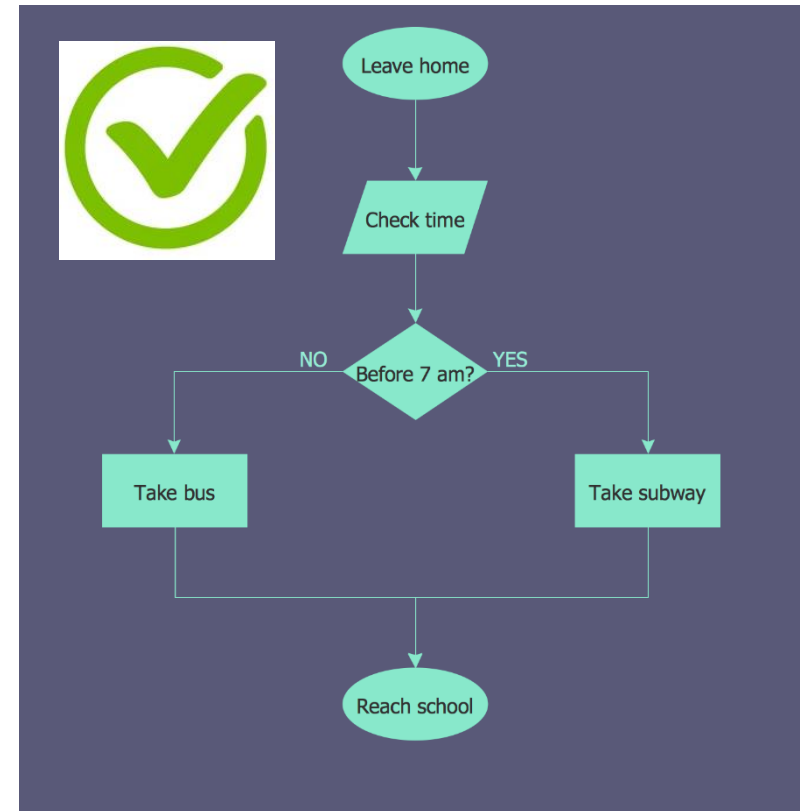
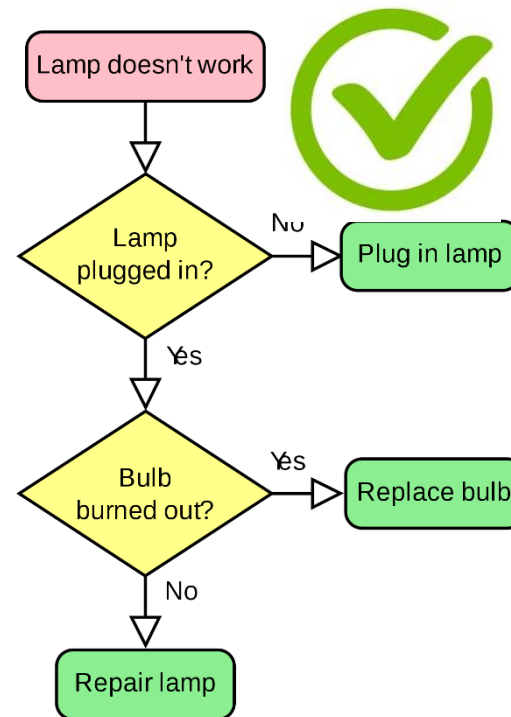
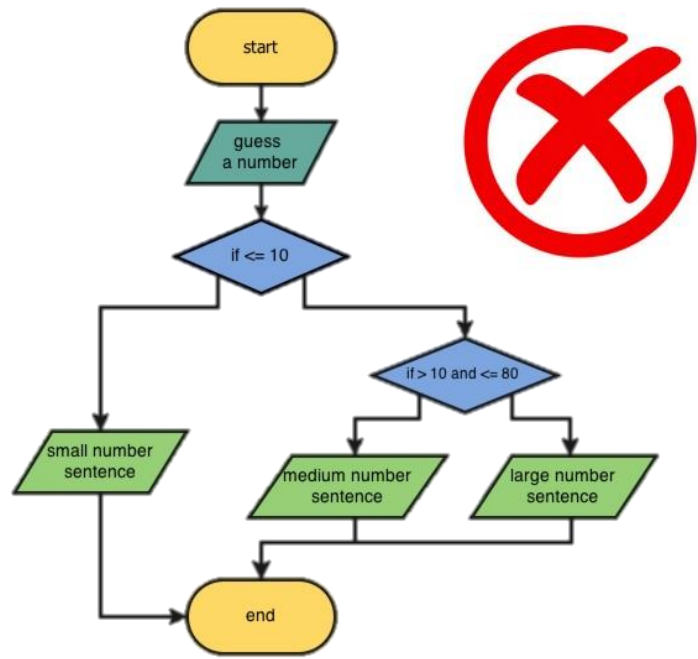
2. Flowchart symbols have an entry point on the top of the symbol with no other entry points.

The exit point for all flowchart symbols is on the bottom except for the Decision symbol.



General Rules in Flowcharting

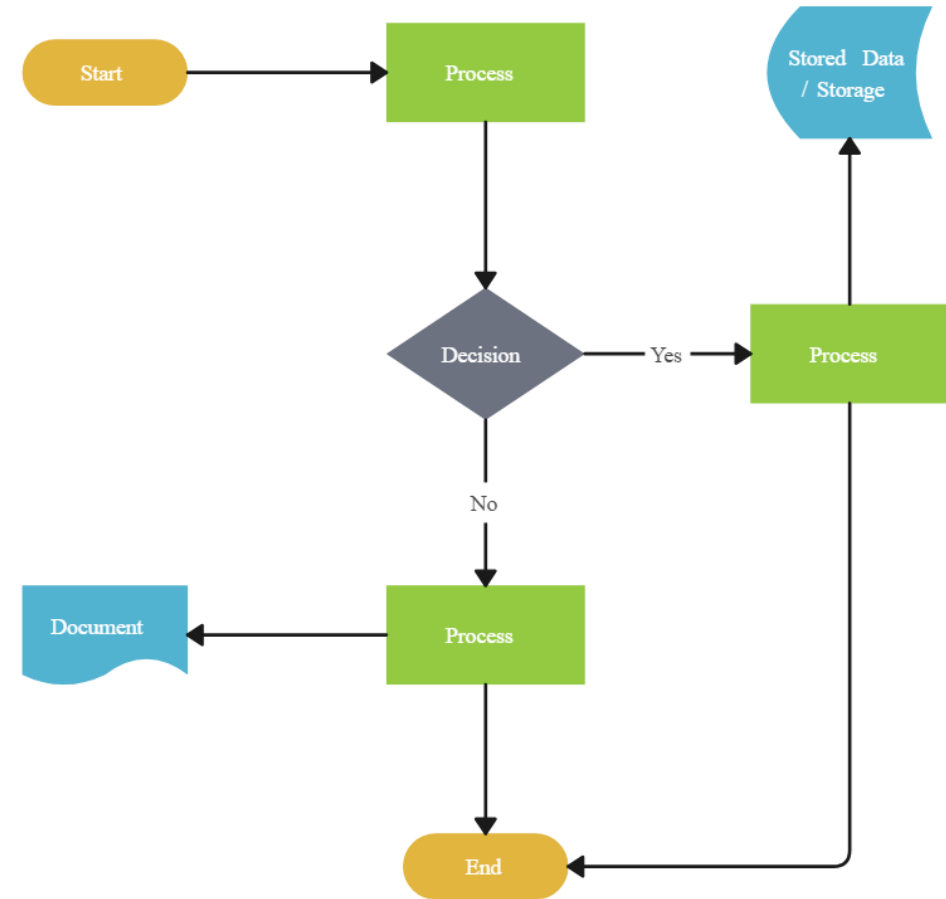
3. The Decision symbol has two exit points; these can be on the sides or the bottom and one side.



General Rules in Flowcharting

4. Generally a flowchart will flow from top to bottom.

However, an upward flow can be shown as long as it does not exceed 3 symbols.

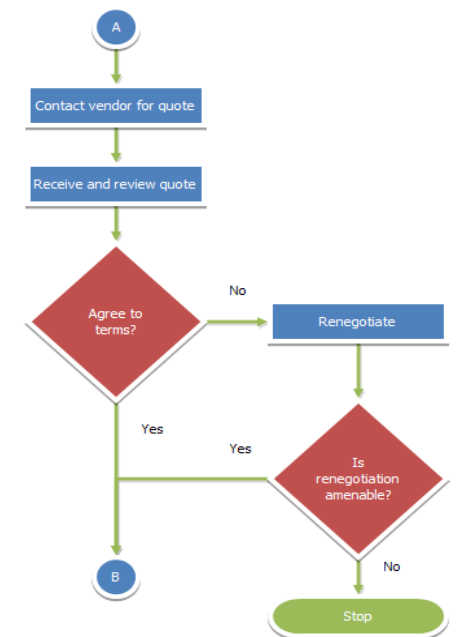
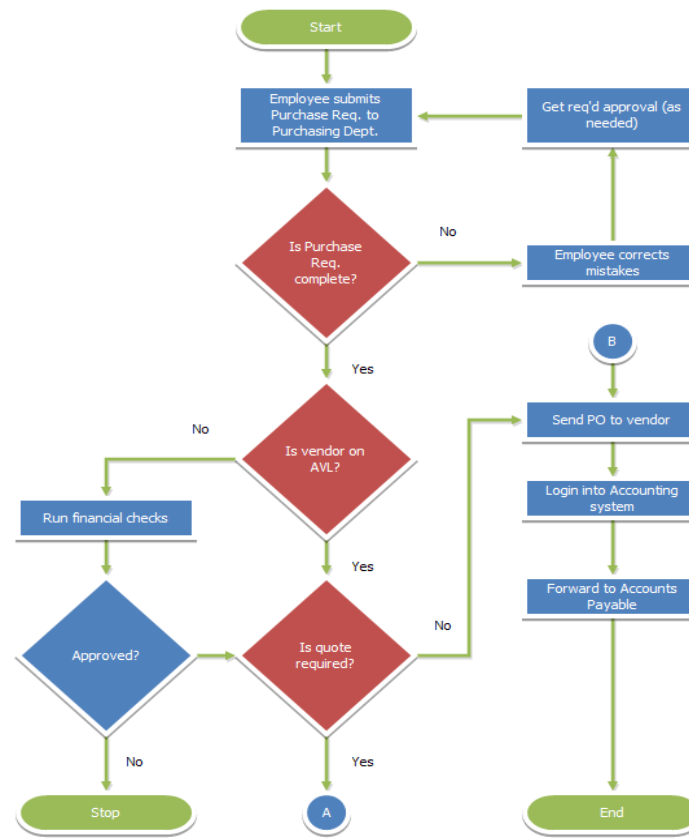


General Rules in Flowcharting

5. Connectors are used to connect breaks in the flowchart.

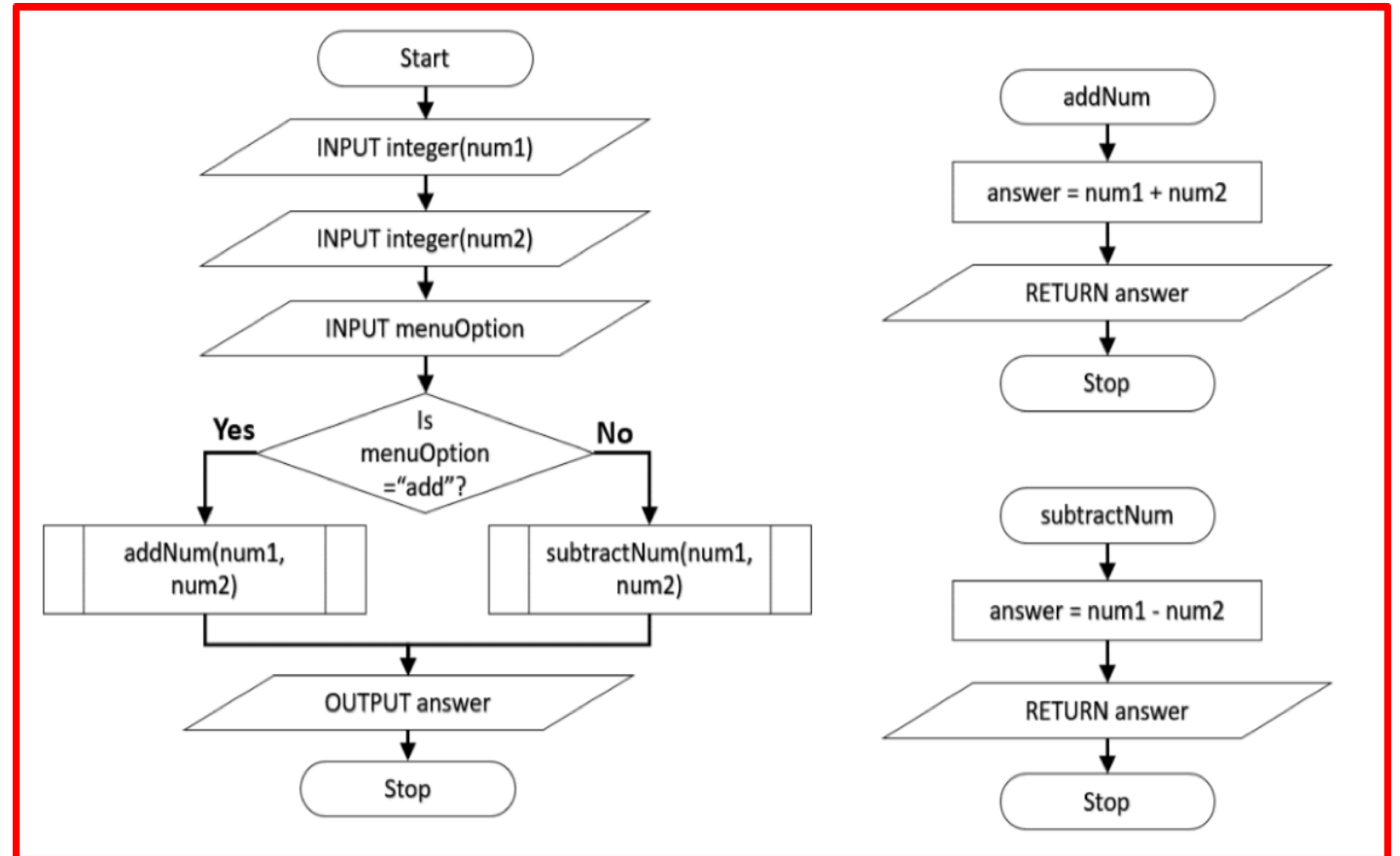
Examples are:

- From one page to another page.
- From the bottom of the page to the top of the same page.
- An upward flow of more than 3 symbols



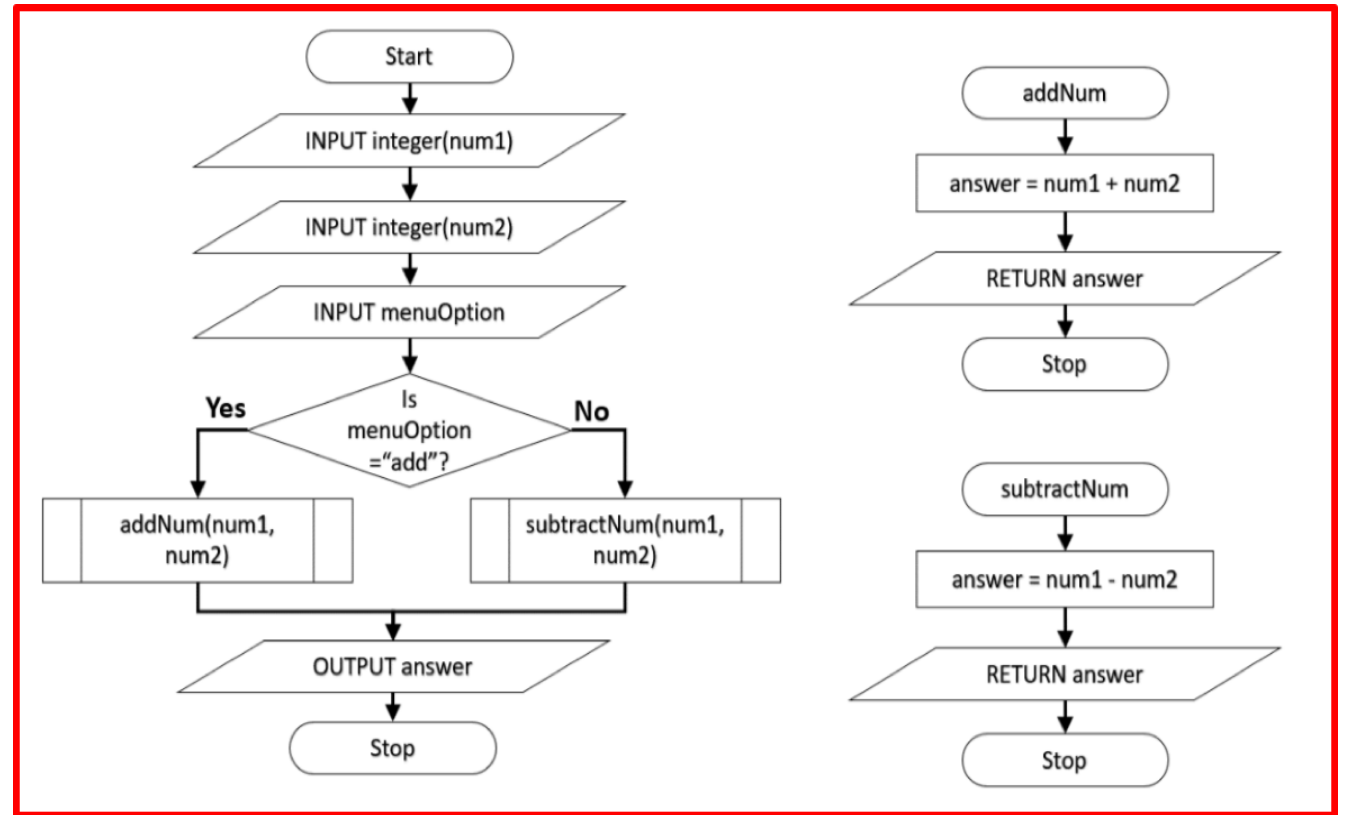
General Rules in Flowcharting

6. Subroutines and Interrupt programs have their own and independent flowcharts.



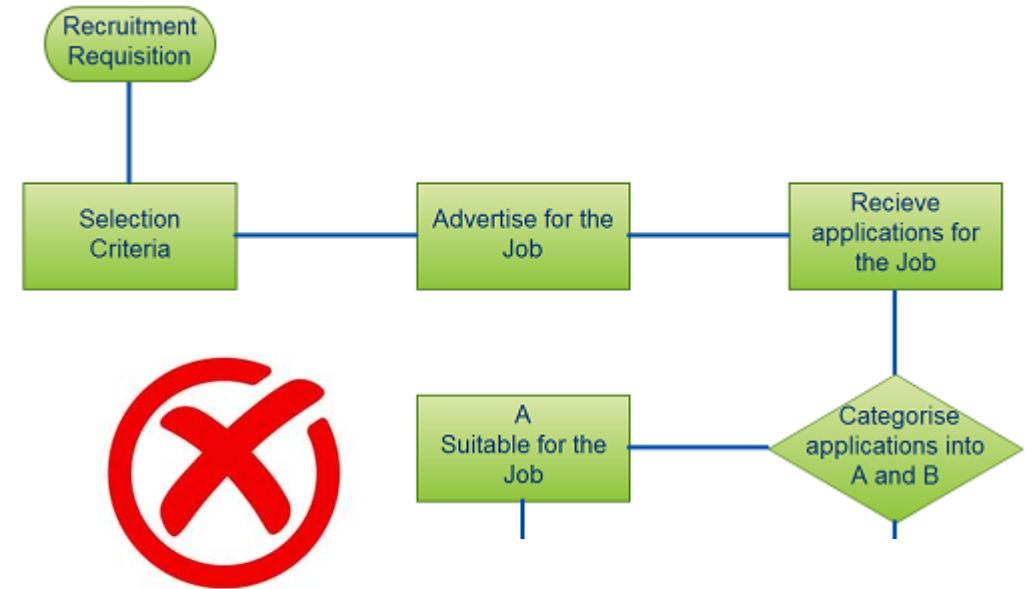
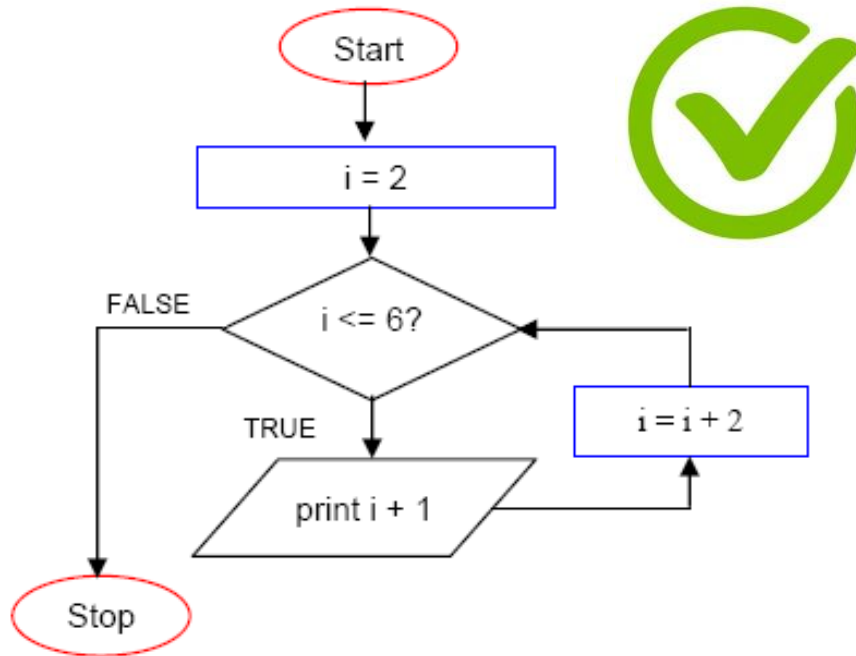
General Rules in Flowcharting

7. All flow charts start with a Terminal or Predefined Process (for interrupt programs or subroutines) symbol.



General Rules in Flowcharting

8. All flowcharts end with a terminal or a continuous loop.

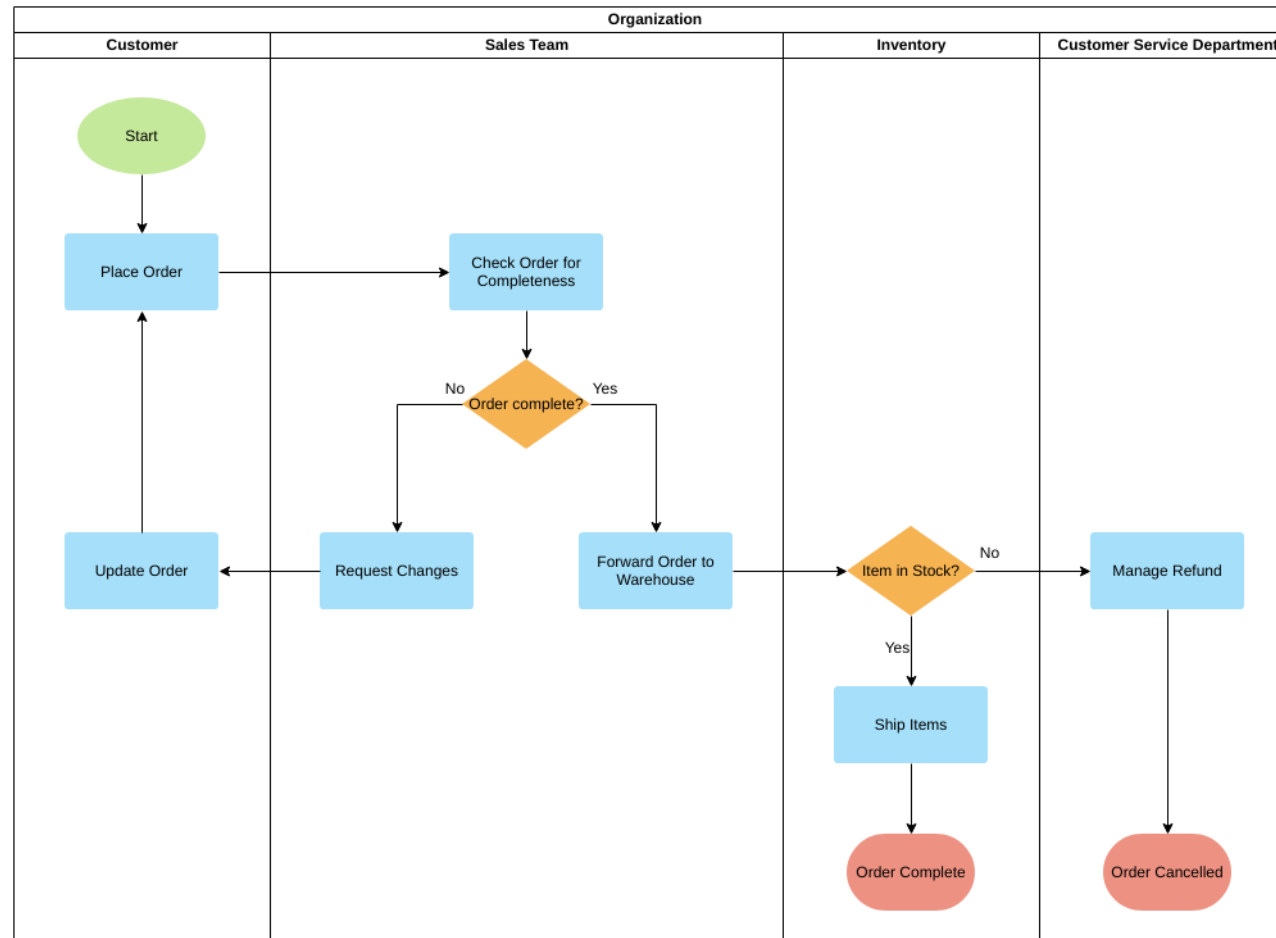


Types of Flowcharts

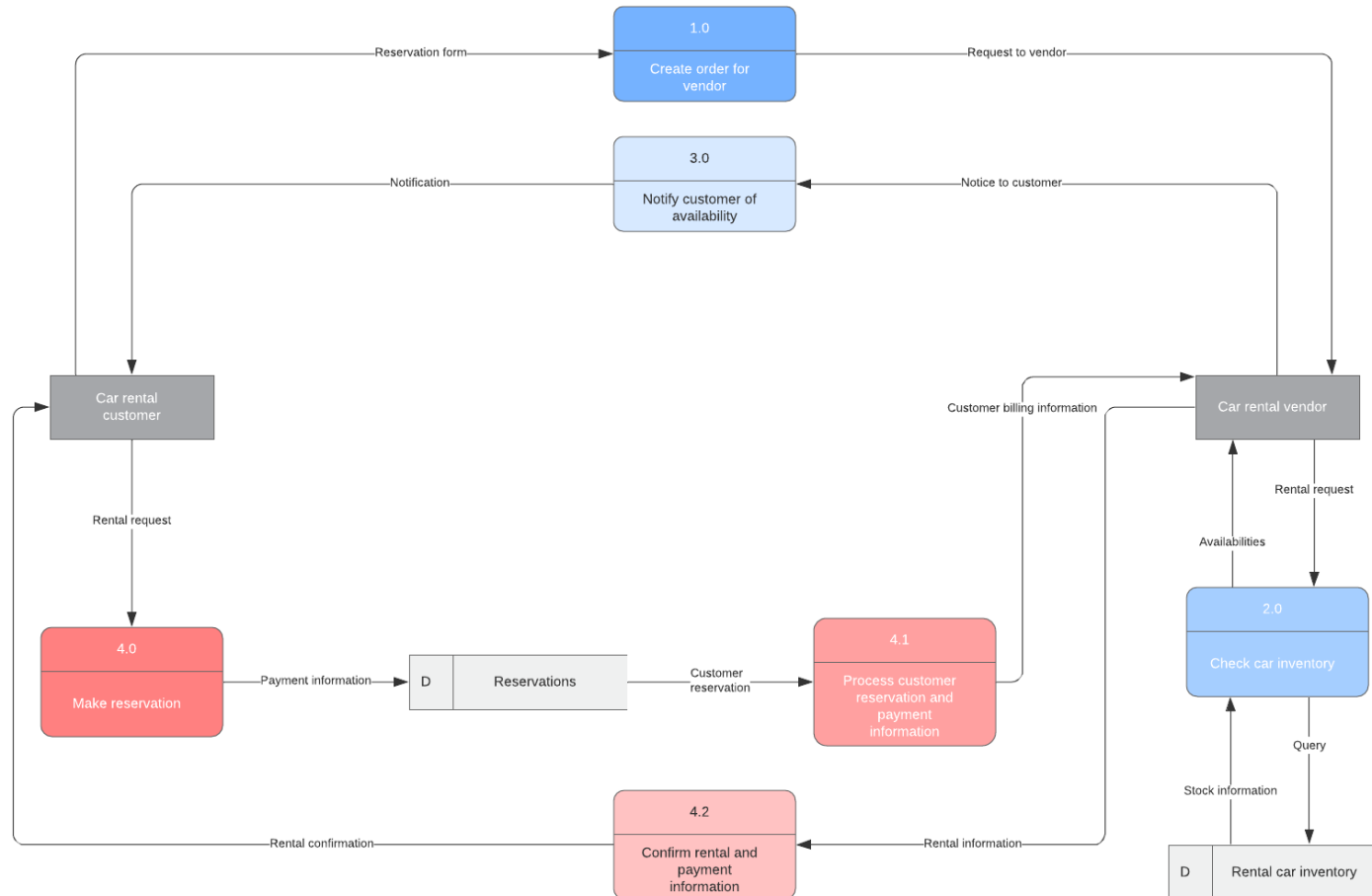
There are a wide variety of flowchart types. Here are just a few of the more commonly used ones.

- Swimlane flowcharts
- Data flow diagrams
- Influence diagrams
- Workflow diagrams
- Process flow diagrams

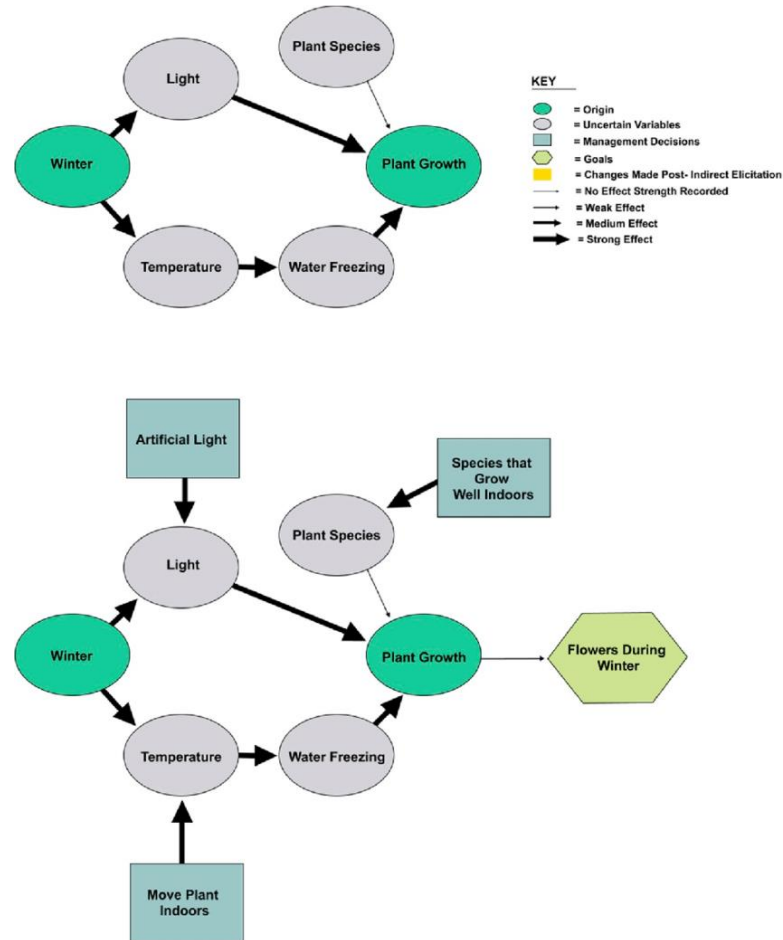
Swimlane Flowcharts



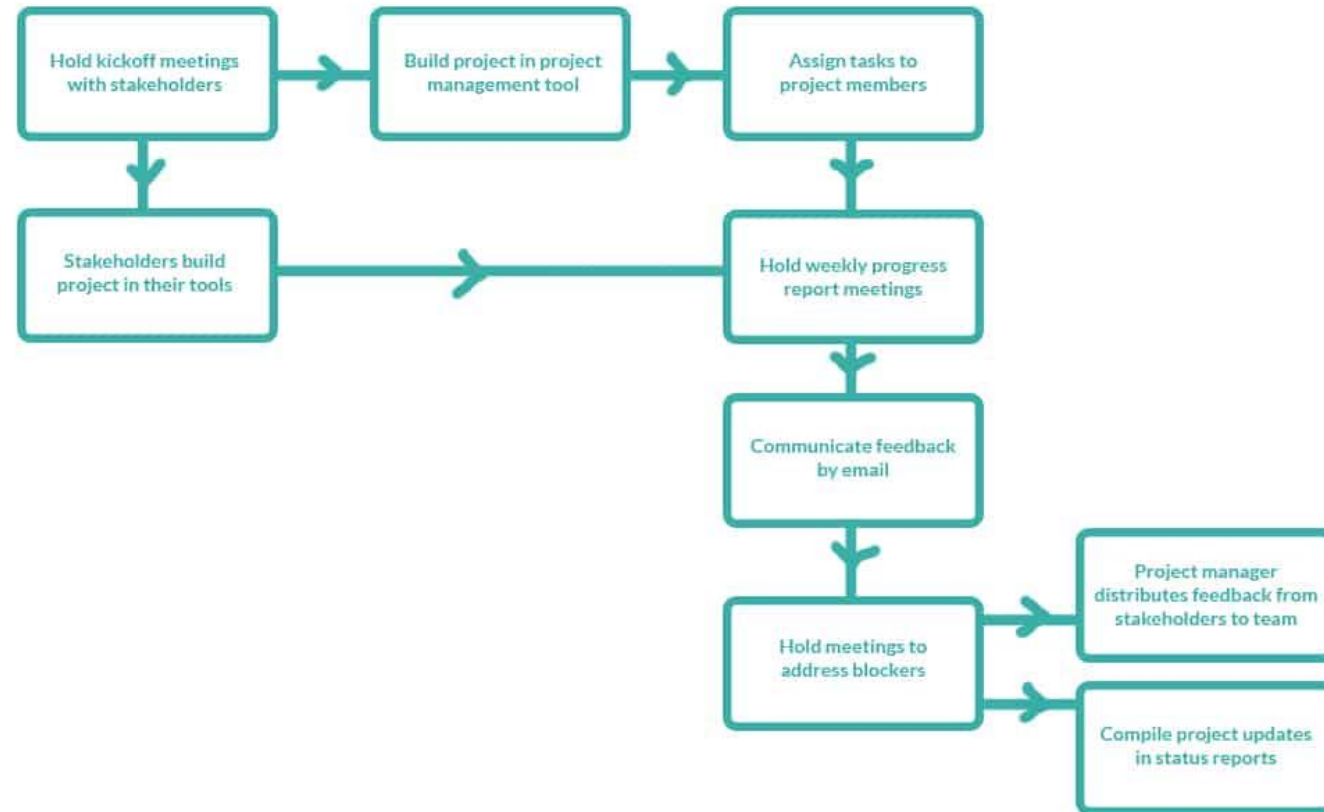
Data Flow Diagram



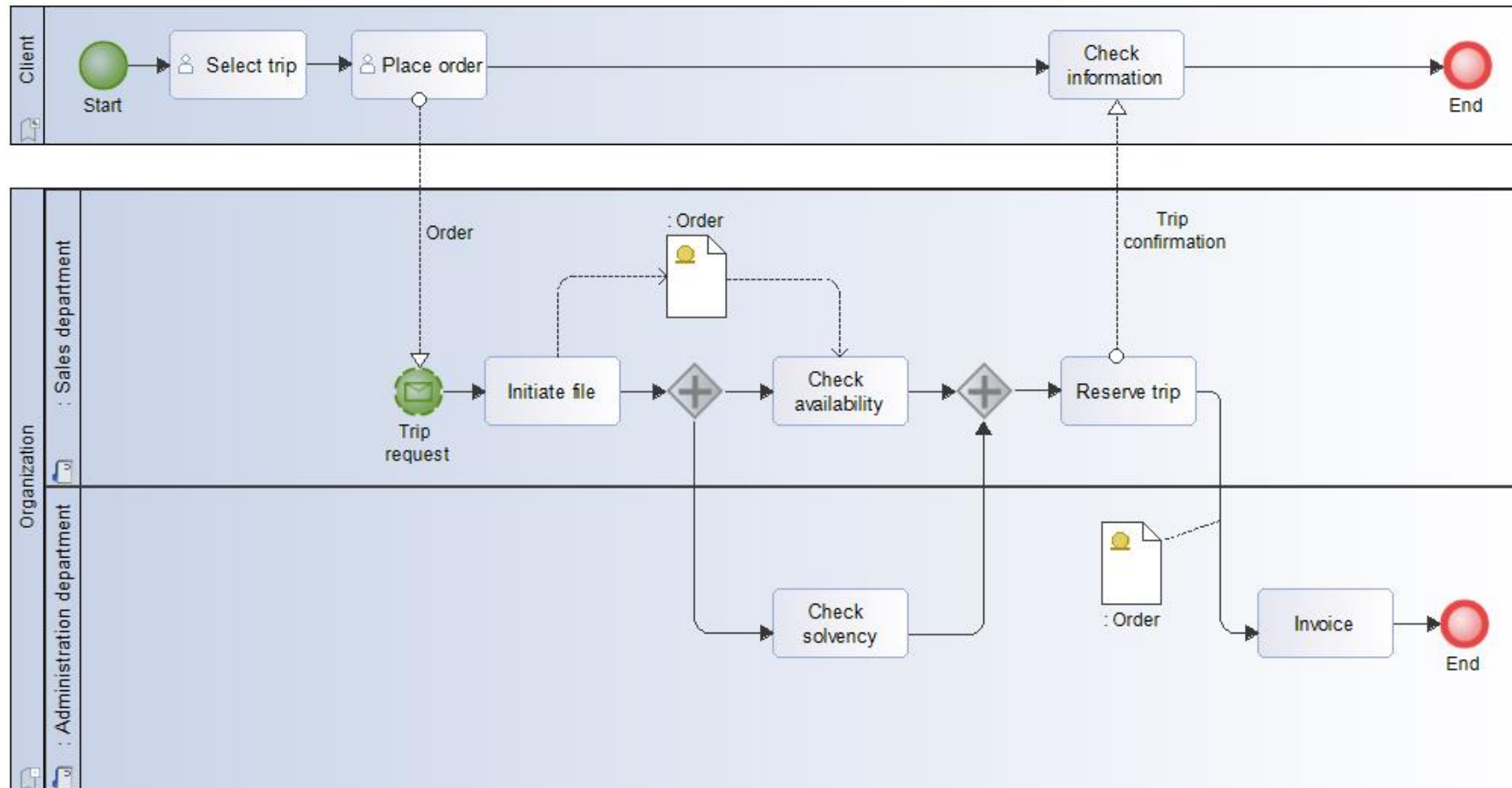
Influence Diagrams



Workflow Diagrams



Process Flow Diagrams



How to plan and draw a basic flowchart

1. Define your purpose and scope.
2. Identify the tasks in chronological order.
3. Organize the steps.
4. Draw your chart.
5. Confirm your flowchart.

Uses of Flowcharts

- When designing and planning a process
- To study a process for better understanding
- To communicate a process
- In computer programming
- In learning environments
- Streamlining manufacturing
- Charting success strategies in Marketing teams



Uses of Flowcharts

- Project planning
- Program or system design
- Map computer algorithms
- Plan coursework and academic requirements
- Show registration flows.
- Represent an employee's tasks or daily routine.
- Understand the paths that users take on a website or in a store.
- Illustrate the manufacturing process from beginning to end.



Advantages of using Flowcharts

- Visual clarity
- Improved communication
- Proper documentation
- Effective analysis
- Efficient Program Maintenance
- Flowcharts act as a guide for blueprint during program designed
- Coding of the Program

Disadvantages of using Flowcharts

- Difficulty in presenting complex programs and tasks
- No scope for alteration or modification
- Reproduction becomes a problem
- It's a time-consuming process
- Difficult to understand for people who don't know flowchart symbols

Flowchart Example

Draw a flowchart to log in to Facebook account.

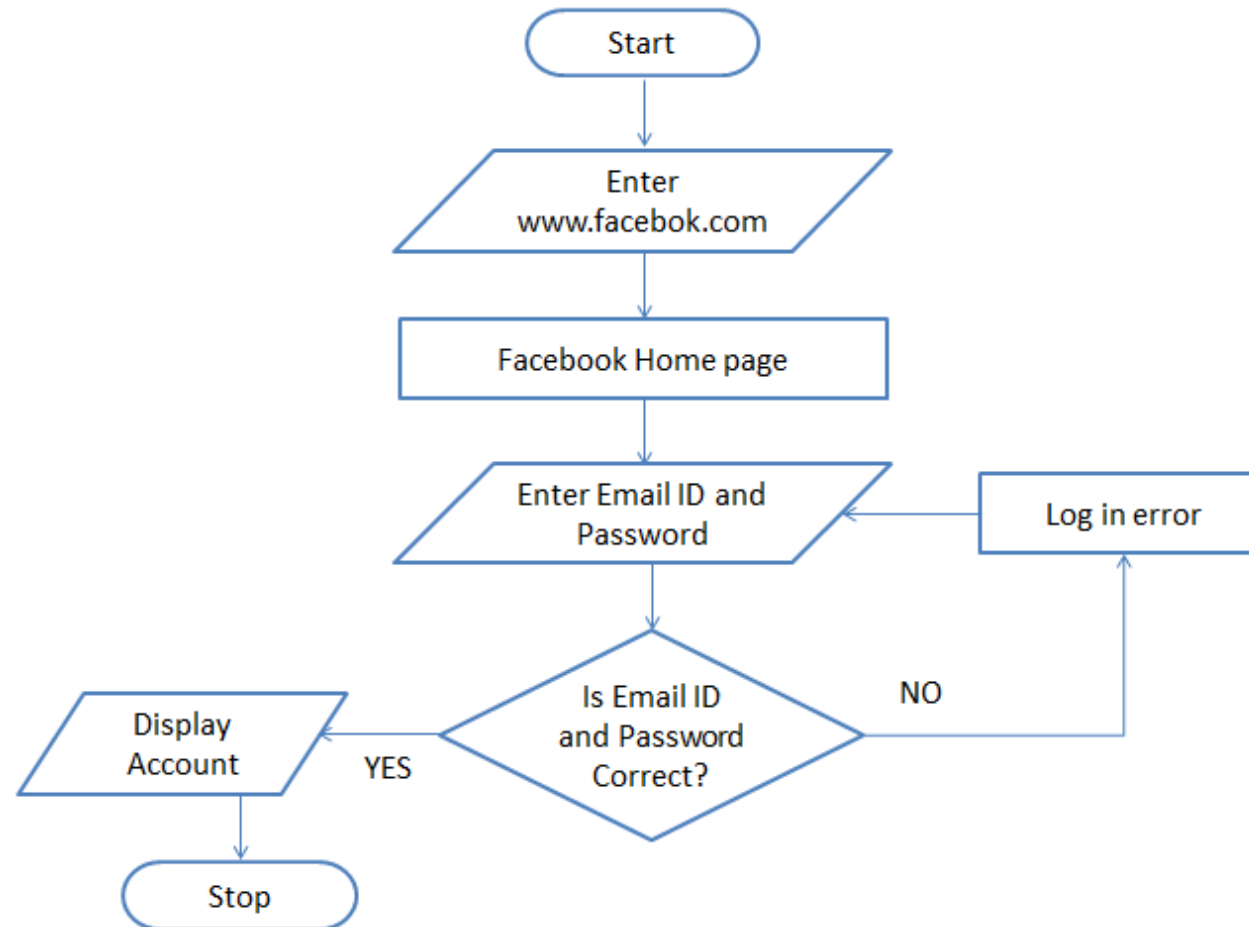
- Draw a flowchart for this problem using your imagination.
- To log in to Facebook account, first enter the Facebook URL www.facebook.com in the browser like Google Chrome, Firefox, Safari, Internet Explorer etc.
- This request is sent to the Facebook server and it responds by sending us the home page of Facebook.
- Next, enter your registered Email ID and Password and click the Login button.
- Then your login credential is checked. If it is correct, it will show our profile.
- On the other hand, if the login credential is wrong then an error occurs and prompted to re-enter our Email ID and Password.

Flowchart Example

Algorithm

1. Enter www.facebook.com in your browser. (I/O)
2. Facebook Home page loads (PROCESS)
3. Enter your Email ID and Password (I/O)
4. Is Email ID and Password Valid (DECISION)
5. if NO then
6. Log in error (PROCESS)
7. go to step 3
8. Else
9. Display Facebook Account (I/O)

Flowchart Example



Flowchart Example

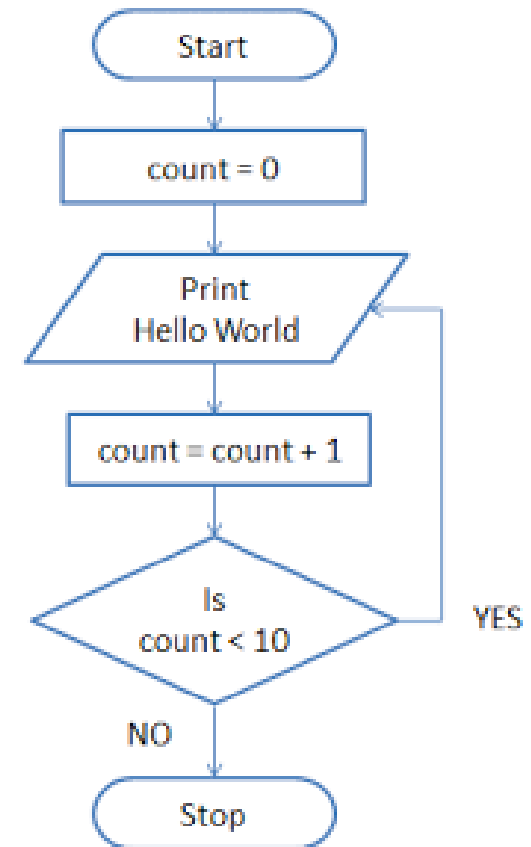
Print Hello World 10 times

- This problem is also solved using the loop concept.
- We take a variable count and set it to zero.
- Then we print "Hello World" and increment count by 1.
- Next we check if count is less than 10. If this is TRUE then we again print "Hello World" and increment the variable count.
- On the other hand if the condition is FALSE then we will stop.

Flowchart Example

Algorithm

1. Initialize count = 0 (PROCESS)
2. Print Hello World (I/O)
3. Increment count by 1 (PROCESS)
4. Is count < 10 (DECISION)
5. if YES go to step 2
6. else Stop

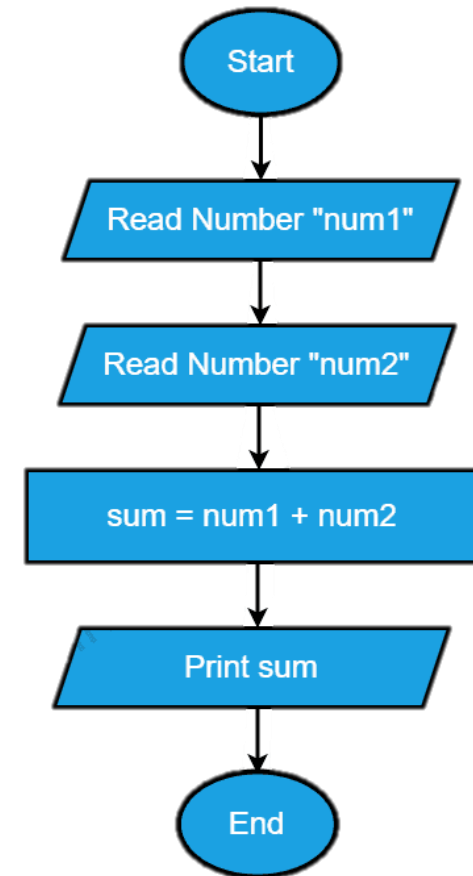


Flowchart Example

Add 10 and 20

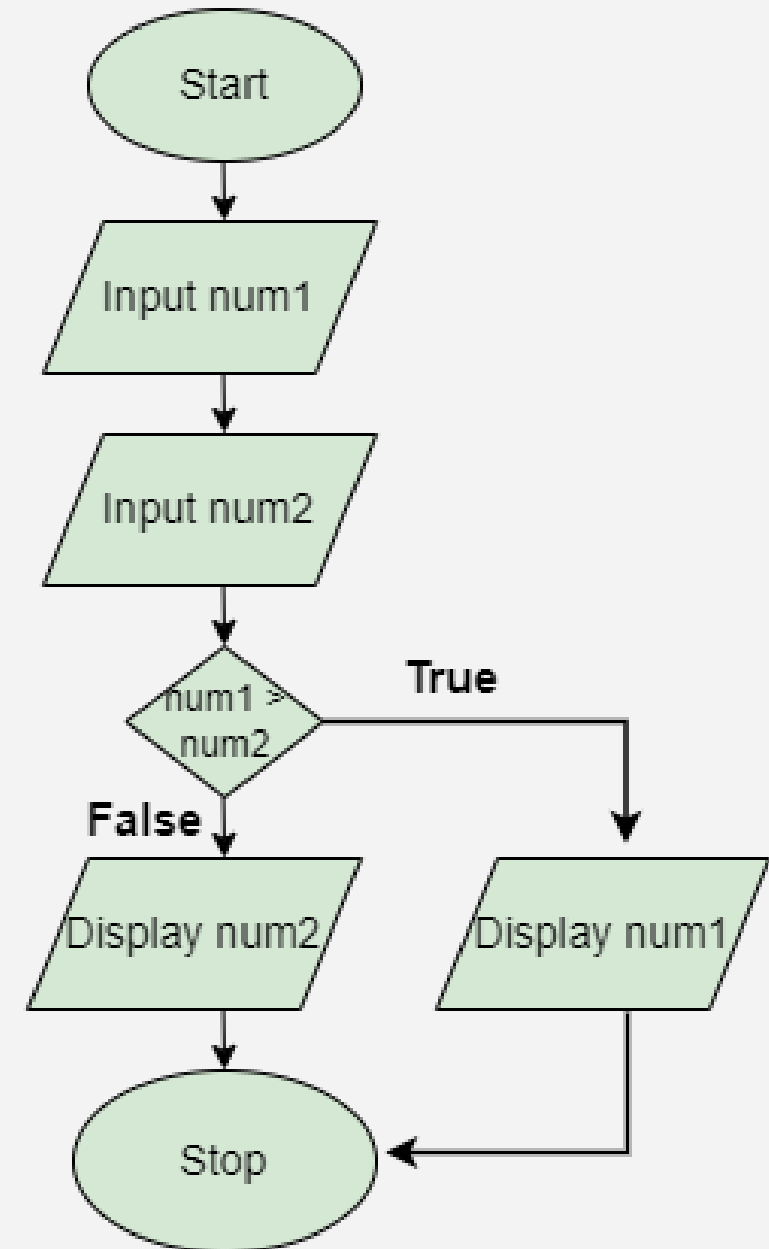
Algorithm

1. Initialize sum = 0 (PROCESS)
2. Enter the numbers (I/O)
3. Add them and store the result in sum (PROCESS)
4. Print sum (I/O)



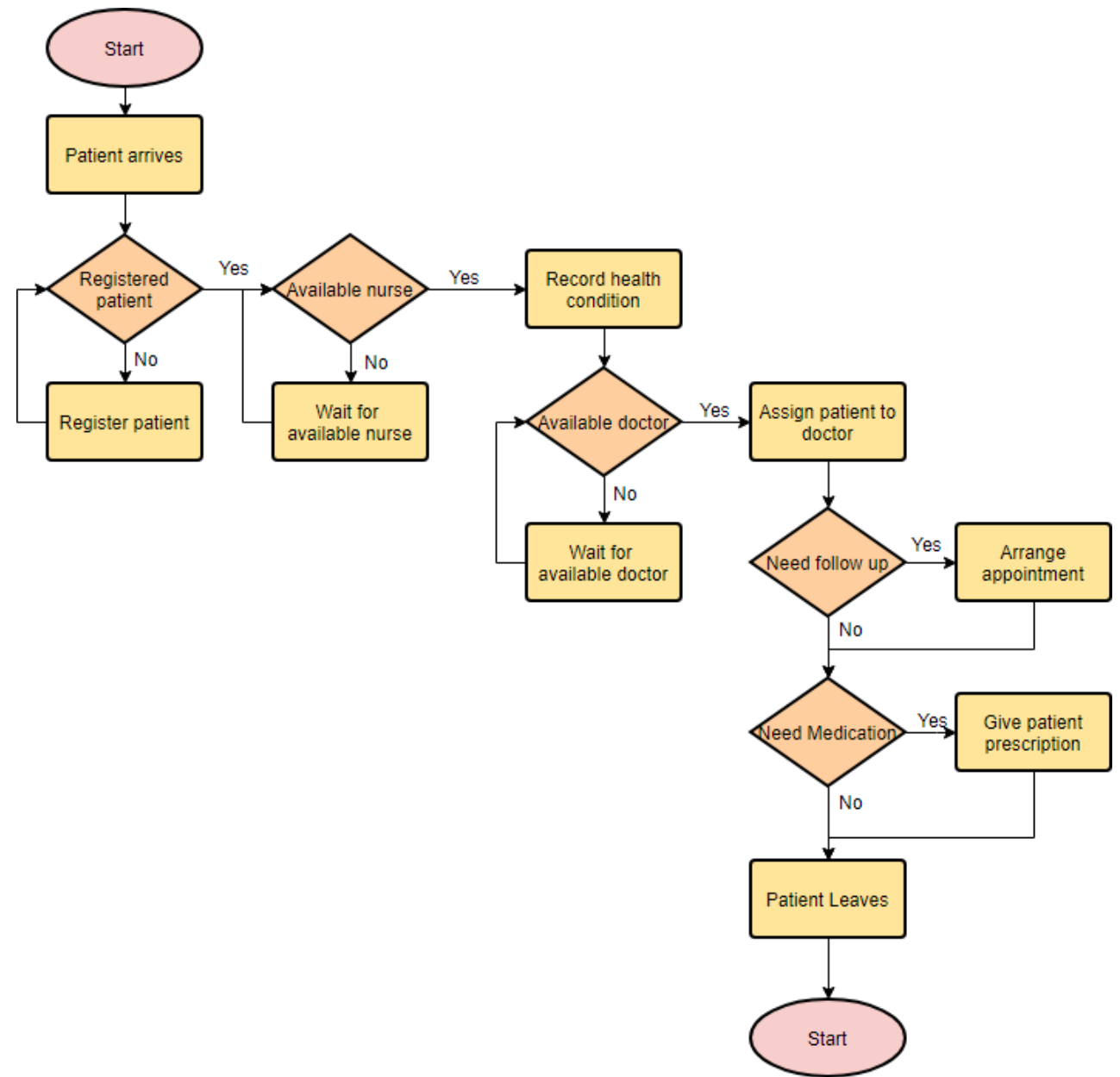
Flowchart Example

Input two numbers from the user and display the largest of two numbers



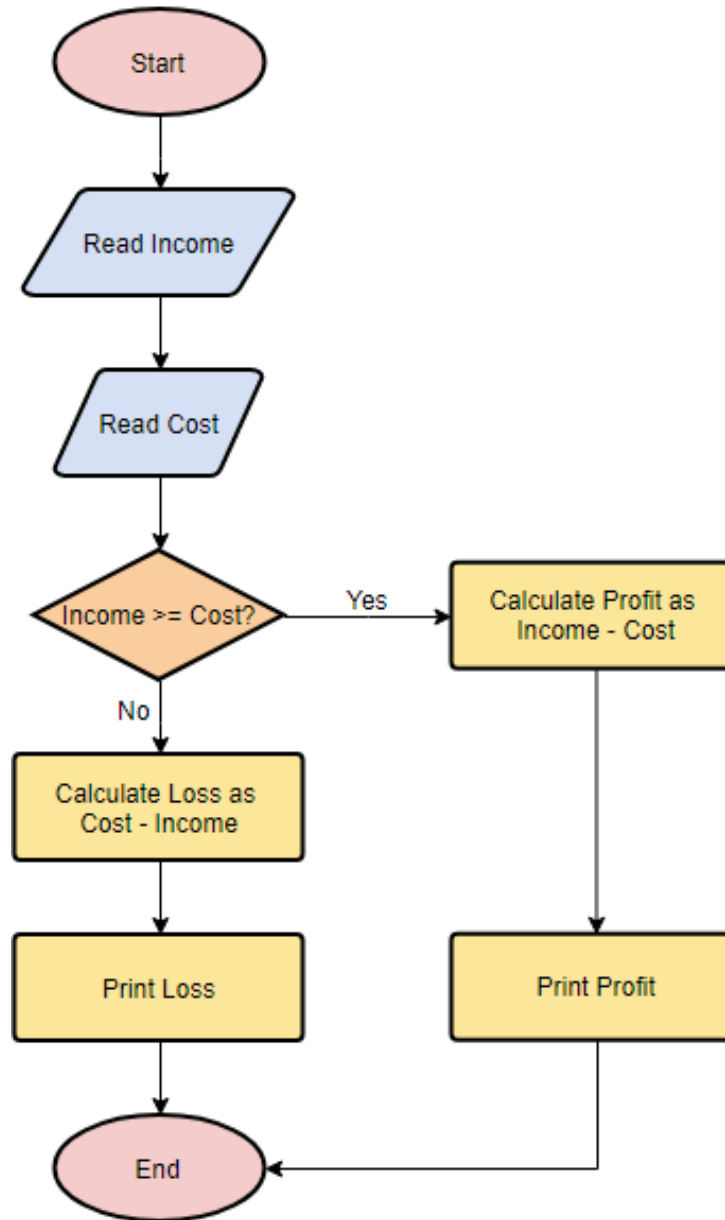
Flowchart Example

Process of medical center



Flowchart Example

Calculate Profit and Loss



Find the profit/loss when
income = 1,000, cost = 800



Thank You
